

The objective of this study was to evaluate the effectiveness of YOLOv8-based object detection models and semantic segmentation architectures for automated road damage analysis.

1. Detection Experiments

Three YOLOv8 variants were trained and evaluated on the RDD2022 dataset.

Baseline Detection Performance

Model	Precision	Recall	mAP@50	mAP@50-95
YOLOv8n	0.579	0.516	0.539	0.285
YOLOv8s	0.626	0.553	0.592	0.317
YOLOv8m	0.677	0.590	0.633	0.344

Observation

YOLOv8m achieved the highest overall detection performance across all evaluation metrics.

2. SAHI Evaluation

SAHI (Slicing Aided Hyper Inference) was applied to improve detection of small road defects.

Detection Count Comparison

Model	Baseline Detections	SAHI Detections	Improvement
YOLOv8n	3773	4564	20.96%

YOLOv8s	4237	5475	29.22%
YOLOv8m	4551	5321	16.92%

Observation

SAHI consistently improved detection performance across all YOLOv8 variants, with the largest gain observed for YOLOv8s. But, it increased the false positive rate, classifying tree branches as alligator cracks for example.

3. Segmentation Experiments

Crack segmentation experiments were conducted using the Crack500 dataset.

Segmentation Performance

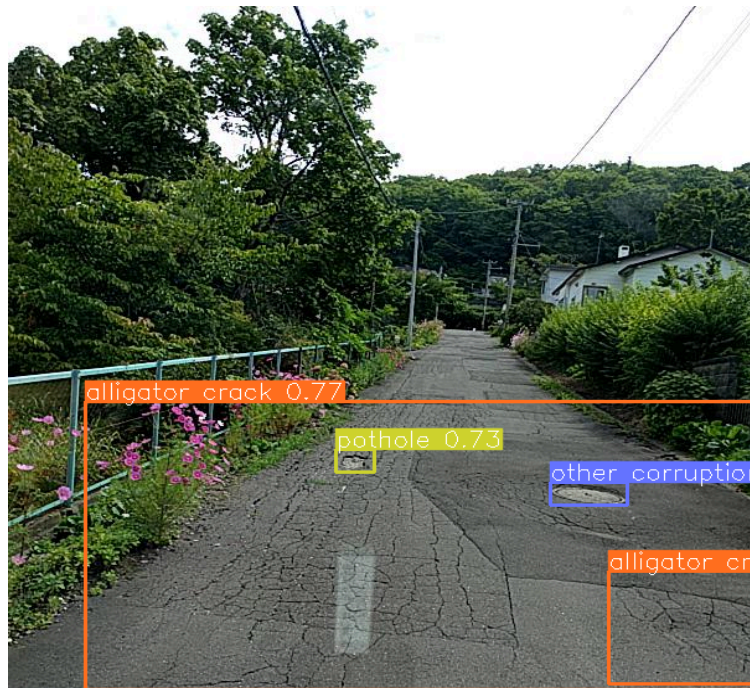
Model	Dice Score	IoU Score
U-Net	0.7533	0.6118
DeepLabV3+	0.7389	0.5931

Observation

U-Net achieved the best segmentation performance and was selected as the final segmentation model. DeepLabV3+ struggled with the finer cracks in the image, which the U-Net was able to account for.

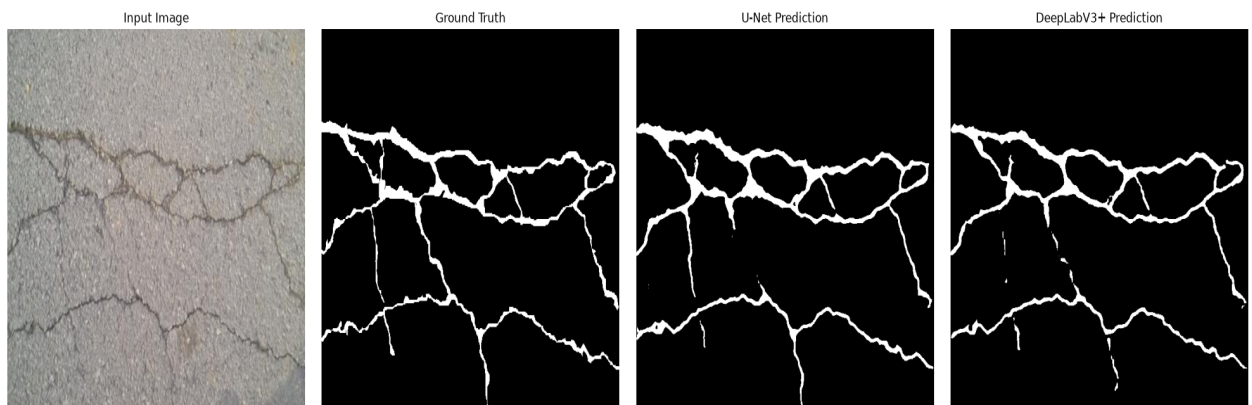
4. Qualitative Results

Successful Detection Example



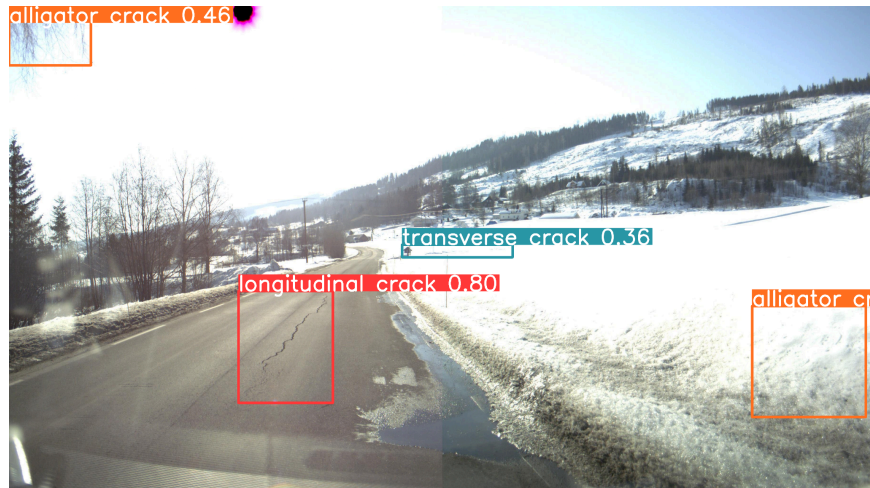
Example showing successful road damage detection using YOLOv8 + SAHI.

Successful Segmentation Example



Example showing crack segmentation prediction compared with ground truth.

Failure Case Example



Example showing false positive detections caused by vegetation, shadows, snow, or road-side textures.

5. Key Findings

- YOLOv8m achieved the best baseline detection performance.
- SAHI improved the number of detected road defects across all YOLOv8 models.
- YOLOv8s + SAHI achieved the highest improvement (+29.22%).
- U-Net outperformed DeepLabV3+ on the Crack500 dataset.
- Common failure cases include vegetation, shadows, snow-covered regions, and high-contrast road textures.